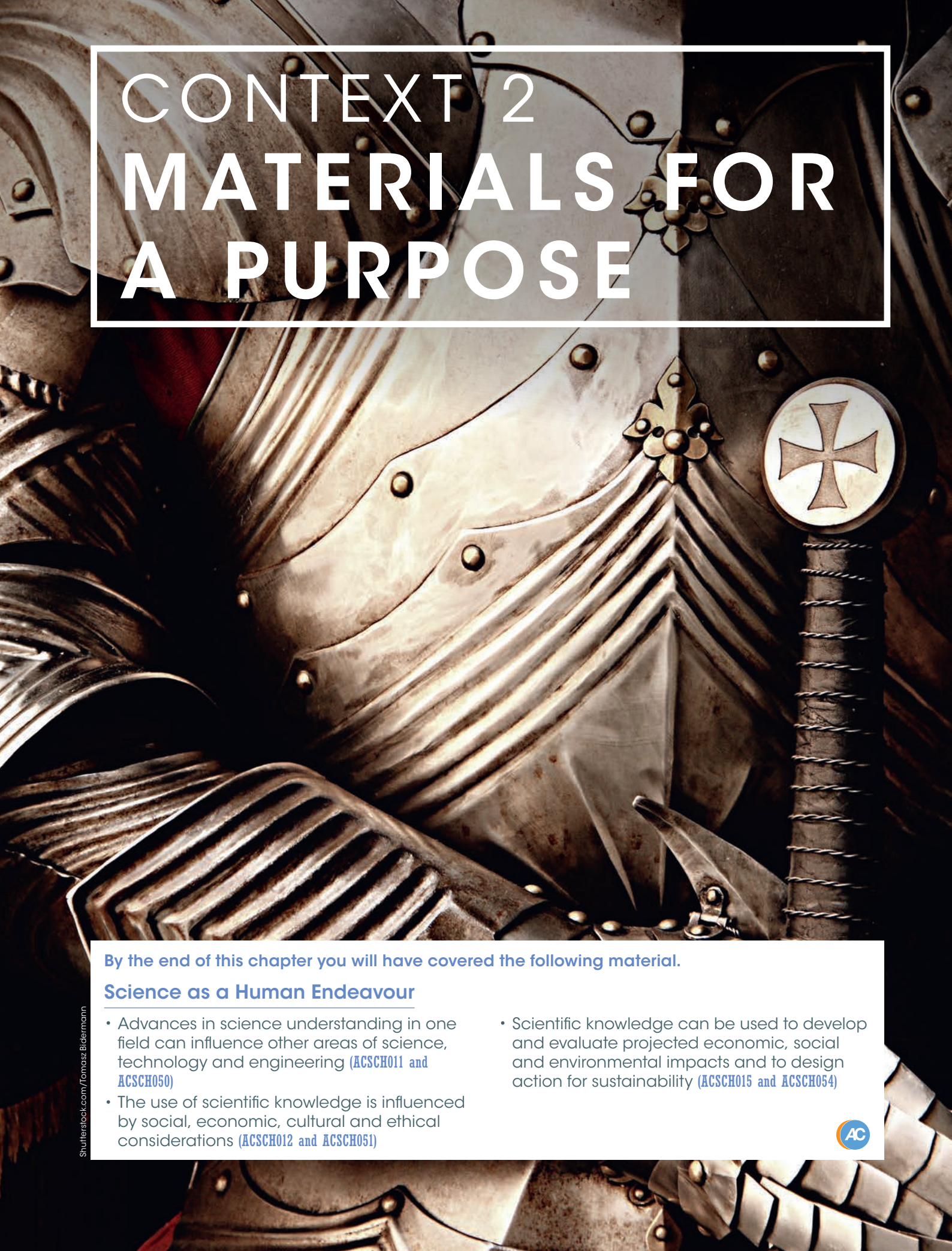




CONTEXT 2

MATERIALS FOR

A PURPOSE

By the end of this chapter you will have covered the following material.

Science as a Human Endeavour

- Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)
- The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)
- Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability (ACSCH015 and ACSCH054)

Contextual story – Materials for a purpose

- History – humans have always tried to make use of the materials around them
- Why most of Earth's materials are not found as elements
- Natural materials
- Other natural materials that are changed to useful materials
- Finding new materials we didn't know existed
- Looking after our resources

Chemical understanding

- Bonding and properties of materials
- Writing formulas
- Pure substances and mixtures
- Chemical and physical properties
- Chemical reactions
- Balancing chemical equations
- Percentage composition
- Ionic bonding and naming
- Reactivity
- Covalent bonding and naming
- Hydrocarbons
- Methods of separation of mixtures
- Exothermic and endothermic reactions
- Mass and moles, basic stoichiometry and law of conservation of mass
- Thermochemistry
- Biofuels and fossil fuels

Investigations

- Context 2**
 - Investigations 2.1 and 2.2
- Chapter 2**
 - Experiment 2.1
 - Investigation 2.1
- Chapter 3**
 - Experiments 3.1 and 3.2
 - Activity 3.1
- Chapter 4**
 - Investigations 4.1 and 4.2
- Chapter 5**
 - Experiment 5.1

Science as a Human Endeavour (SHE)

- Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)
- The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)
- Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability predictions (ACSCH015 and ACSCH054)

Science Understanding (SU)

- Materials are either pure substances with distinct measurable properties (for example, melting and boiling point, reactivity, strength, density) or mixtures with properties dependent on the identity and relative amounts of the substances that make up the mixture (ACSCH025)
- Differences in the properties of substances in a mixture, such as particle size, solubility, magnetism, density, electrostatic attraction, melting point and boiling point, can be used to separate them (ACSCH026)
- The type of bonding within substances explains their physical properties, including melting and boiling point, conductivity of both electricity and heat, strength and hardness (ACSCH027)
- Chemical bonds are caused by electrostatic attractions that arise because of the sharing or transfer of electrons between participating atoms; the valency is a measure of the number of bonds that an atom can form (ACSCH029)
- Ions are atoms or groups of atoms that are electrically charged due to an imbalance in the number of electrons and protons; ions are represented by formulae which include the number of constituent atoms and the charge of the ion (for example, O^{2-} , SO_4^{2-}) (ACSCH030)
- The properties of ionic compounds (for example, high melting point, brittleness, ability to conduct electricity when liquid or in solution) are explained by modelling ionic bonding as ions arranged in a crystalline lattice structure with forces of attraction between oppositely charged ions (ACSCH031)
- Covalent substances are modelled as molecules or covalent networks that comprise atoms which share electrons, resulting in electrostatic forces of attraction between electrons and the nucleus of more than one atom (ACSCH033)
- Carbon forms hydrocarbon compounds, including alkanes and alkenes, with different chemical properties that are influenced by the nature of the bonding within the molecules (ACSCH035)
- All chemical reactions involve the creation of new substances and associated energy transformations, commonly observable as changes in the temperature of the surroundings and/or the emission of light (ACSCH036)
- Endothermic and exothermic reactions can be explained in terms of the law of conservation of energy and the breaking and reforming of bonds; heat energy released or absorbed can be represented in thermochemical equations (ACSCH037)
- Fuels, including fossil fuels and biofuels, can be compared in terms of their energy output, suitability for purpose, and the nature of products of combustion (ACSCH038)
- A mole is a precisely defined quantity of matter equal to Avogadro's number of particles; the mole concept and the law of conservation of mass can be used to calculate the mass of reactants and products in a chemical reaction (ACSCH039)

Science Inquiry Skills (SIS)

- Identify, research and refine questions for investigation; propose hypotheses; and predict possible outcomes (ACSCH001)
- Design investigations, including the procedure/s to be followed, the materials required, and the type and amount of primary and/or secondary data to be collected; conduct risk assessments; and consider research ethics (ACSCH002 and ACSCH041)
- Conduct investigations, including the use of devices to accurately measure temperature change and mass, safely, competently and methodically for the collection of valid and reliable data (ACSCH003)
- Represent data in meaningful and useful ways, including using appropriate graphic representations and correct units and symbols; organise and process data to identify trends, patterns and relationships; identify sources of random and systematic error and estimate their effect on measurement results; and select, synthesise and use evidence to make and justify conclusions (ACSCH004)
- Interpret a range of scientific and media texts, and evaluate processes, claims and conclusions by considering the quality of available evidence; and use reasoning to construct scientific arguments (ACSCH005 and ACSCH044)
- Select, construct and use appropriate representations including chemical symbols and formulae, molecular structural formulae, physical and graphical models of structures, chemical equations and thermochemical equations, to communicate conceptual understanding, solve problems and make predictions (ACSCH006)
- Select and use appropriate mathematical representations to solve problems and make predictions, including calculating percentage composition from relative atomic masses and using the mole concept to calculate the mass of reactants and products (ACSCH007)
- Communicate to specific audiences and for specific purposes using language, nomenclature, genres and modes, including scientific reports (ACSCH008 and ACSCH047)



Materials

Humans have always been fascinated with the materials around them. We have tried to understand the properties of the materials and use them for specific purposes. Chemistry involves the study of materials – analysing the composition of materials, determining how elements, compounds and mixtures are structured, and exploring the properties of the materials. When we understand the structure and properties of materials, both physical and chemical, we can see the possibilities for their use. Chemistry also investigates the way that substances react to form new materials. These materials may replace other materials because they have properties better suited to a particular use or they may be used for a completely new purpose.

From the earliest civilisations of the Stone Age through the Bronze Age and Iron Age to more recent times, humans have investigated materials. We have utilised the most common and most easily accessible and separated materials, from stone to metals to crude oil. We have found materials that have the properties we need for specific uses. As we have come to better understand the impact of materials and their use, the focus has turned to ways of minimising this impact. We are trying to develop materials that have a reduced **environmental footprint** and develop ways to reuse materials to lessen resource use.



To revise classification and properties of materials, refer to Chemistry sections 2.1 and 2.2 on pages 140 and 143.



To revise mixture separation techniques, refer to Chemistry section 2.3 on page 150.

History – making use of available materials

Stone Age

The Stone Age occurred from the time of the earliest humans to about 3000 BCE. The earliest humans used stone. Initially, they used a piece of stone to smash against another piece of stone to grind ingredients for food. The stones worked like a mortar and pestle (Figure 2.1).

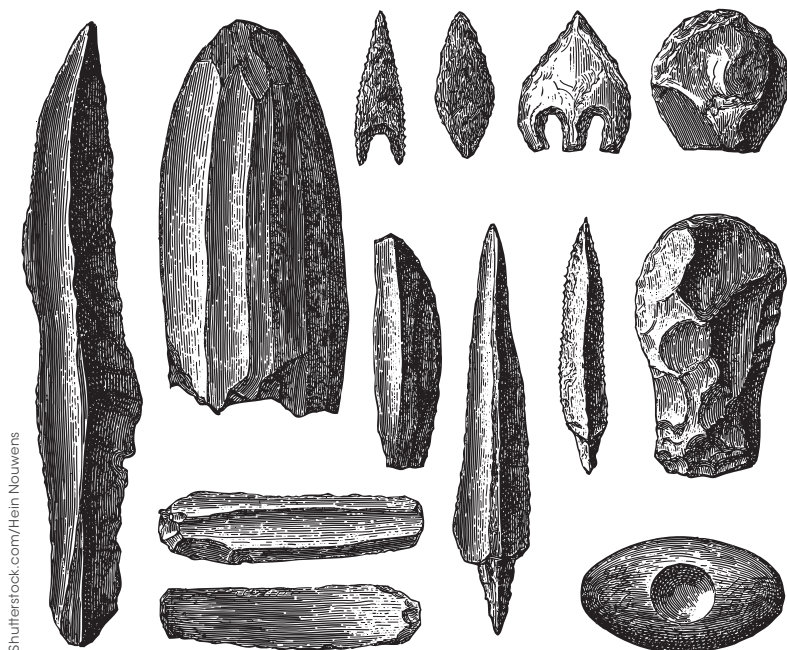
However, over time people realised that different stones have different properties. They found that some stones were very hard, while others could be shaped to produce useful implements and more advanced tools, such as arrowheads for spears that were formed by flaking obsidian (a volcanic rock) or flint.



Shutterstock.com/bonchan

▲ Figure 2.1

A mortar and pestle is used for crushing and grinding substances. The pestle grinds the material in the bowl-shaped mortar.



Shutterstock.com/Hein Nouwers

▲ Figure 2.2

Stone Age tools



INDIGENOUS CAVE PAINTINGS

See *COSMOS* for an article on Northern Territory cave paintings believed to be the oldest in the world.

However, gradually people found other uses for Earth's resources. They used clay to produce the first pottery vessels for holding food and drink. They exploited different-coloured pigments in different ochres and materials to decorate their pottery. Red ochre was widely used for over 200 000 years. It is derived from the mineral haematite (iron(III) oxide, Fe_2O_3), a compound common in many regions of the world. They also used charcoal (the element, carbon) from their fires for the colour black. Both of these materials are powders and will not adhere to pottery or cave walls. So, they mixed eggs or tree sap with the pigments. The sap and eggs act as binders to enable the coloured powders to bind to the surface.

This mixing of binders and pigments was one of the earliest examples of people mixing materials to change their properties to make them more useful. People used sticks, or sticks with fine grasses attached, for their brushes to apply their pigments to cave walls or pieces of bark. Stone Age people effectively used the materials available to them to create weapons and containers, and for decorating and artwork.



Figure 2.3 ►

This Greek pottery vase is from the Late Neolithic period (5300–4500 BCE) in the Stone Age.

Getty Images/De Agostini/DAGLI ORTI

INVESTIGATION 2.1

CREATING AND USING PIGMENTS

The earliest people used mixtures of pigments with different binders to produce cave paintings and decorate their bodies for ceremonial purposes. In this investigation, you will research and experiment with different pigments and binders to create artworks on different materials such as paper, stone and linen.

What is your aim?

To identify materials that can be used as pigments and binders and determine which binders are best to use on different materials

What will you need?

Decide on the colours you want to use for your artwork. Research simple chemicals that can be used to produce these pigments.

Find out the types of materials on which the earliest people created their art. Research where you can get samples of these materials.

Research chemicals that the earliest people used to bind pigments to the material on which they produced artwork. Two commonly available examples are water and egg.

What are the risks?

Construct a table similar to the one below. Identify specific risks involved in the investigation and ways that you will manage the risks to avoid injuries or damage to equipment. Ask your teacher to check your risk assessment before you proceed.

What are the risks in doing this investigation?	How can you manage these risks to stay safe?
Pigments may stain your skin.	

How will you carry out your investigation?

Describe a method for conducting this investigation. Clearly indicate which binders will be used with which pigments and which materials will be used for each of the pigment-binder combinations.

What results will you collect?

Make sure that you know which results you will be collecting, and how and when you will be collecting them.

Discuss the best way to record your results. Draw a table or other format in readiness.

Discuss how you will best present your results in a visual way so other people can understand them.

What have you found?

Did you use the same binder to hold the pigments on to all of the materials? If not, justify which binders were most useful for each material.

What do you conclude?

Write a short conclusion to your investigation based on your aim and what you have found.

Ideas for improvement or further investigation

List any ideas on how to improve what you did, or areas that could warrant further investigation.

Bronze Age

The Bronze Age occurred from about 3000 BCE to 1000 BCE. During this time people began to use metals, such as copper and bronze – an **alloy** (mixture of a metal and another element).

Gold was the first metal to be used by ancient civilisations because its colour and lustre made it easy to recognise. Frequently, gold deposits were impure and had a high silver content. Hence, the gold alloy, electrum, was more commonly used. These deposits were rare and, therefore, electrum was used for the jewellery of leaders of these ancient civilisations.

Copper was first used about 5000 BCE. The first copper mineral to be used was malachite, $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$. Initially, these earliest metallurgists processed malachite by placing it in a fire. However, this gave a very low yield. Over time, better techniques were developed, such as using stones to pound the copper ore into smaller pieces, which were then washed and placed into a simple clay kiln. A good air supply and charcoal in the kiln were needed to get a good yield of copper. Once extracted, the copper was reheated so that it could be hammered into shape for jewellery and containers.

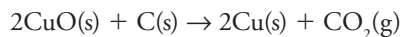


To learn about chemical reactions and balancing chemical equations, refer to Chemistry section 4.1 on page 206.



To learn about the relative masses of substances and percentage composition, refer to Chemistry sections 5.1 and 5.2 on pages 230 and 233.

In the fire, and later in clay kilns, two chemical reactions were occurring. The following equations show the reactants and products for these processes. They also indicate the relationship between the various reactants and products.



As chemistry developed, metallurgists found it useful to determine yields of products through using chemical reactions.

ACTIVITY 2.1

COMPARING COMPOSITION OF COPPER ORES

Aim

To determine the percentage of copper in some copper ores

What to do

- 1 Determine the percentage of copper in each of the following ores.
 - a Malachite, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
 - b Chalcopyrite, CuFeS_2
 - c Cuprite, Cu_2O
 - d Chalcocite, Cu_2S
 - e Tennantite, $\text{Cu}_{12}\text{As}_4\text{S}_{13}$
- 2 Research and discuss why copper was first extracted from malachite even though it does not have the highest percentage of copper.

Over time, malachite sources were used up and so people began to use copper sulfide ores, such as chalcopyrite, CuFeS_2 . These sulfide ores were more difficult to smelt and contained more impurities. So, after processing, the final product was rarely pure copper, but was more likely to be an alloy of copper with a small amount of arsenic, tin, zinc, antimony or nickel. These impurities affected the properties of the alloy.

ACTIVITY 2.2

EXTRACTING COPPER FROM CHALCOPYRITE

Aim

To describe the physical and chemical processes required to extract copper from chalcopyrite

What to do

- 1
 - a Research the physical and chemical processes used in the Bronze Age for extracting copper from chalcopyrite.
 - b Represent these processes using a flowchart.
- 2 Write balanced chemical equations for each chemical reaction involved in the extraction of copper from chalcopyrite.
- 3
 - a Research the processes currently used for the extraction of copper from chalcopyrite.
 - b Add these processes to your flowchart.
- 4 Justify why the process(es) identified in Question 3 could not be used during the Bronze Age.

As time progressed, people extracting the copper realised that by mixing different ores together, they could produce alloys with different compositions and, therefore, different properties – they were practising some of the skills of modern metallurgy.

◀ **Figure 2.4**

a) Bronze Age artefacts in Rustavi Museum, Georgia; b) Bronze ritual food vessel from Zhou Dynasty, China (550–400 BCE)



ACTIVITY 2.3

COMPARING COPPER WITH COPPER ALLOYS

Aim

To compare the properties of copper and copper alloys

What to do

- 1 Identify the main elements in:
 - a bronze.
 - b brass.
- 2 Identify some uses of copper alloys during the Bronze Age.
- 3 Outline some of the health issues faced by metallurgists during the Bronze Age.
- 4 Research the properties and uses of copper and at least four copper alloys.

What did you discover?

- 1 Relate the properties of copper and the copper alloys to their uses.
- 2 Justify why copper alloys are more commonly used than copper.
- 3 Justify why there was a time period called the 'Bronze Age' rather than the 'Copper Age'.
- 4 Explain why the Bronze Age ended.



Iron Age

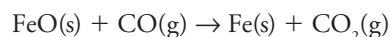
The Iron Age occurred from about 1000 BCE to 500 CE. Although iron-rich rocks are found all over the world, iron was not commonly extracted until about 1000 BCE. Extraction of iron from iron ores required specialised furnaces, which could achieve much higher temperatures than those used for the extraction of copper. The early metallurgists of Europe and Asia had a better understanding of the processes and conditions required for the production of bronze than they did for the production of iron.

Ancient iron artefacts were very precious because iron is generally found in ores and requires a significant investment of time and energy to extract it. Elemental iron was found at only a few sites where meteors had impacted Earth's surface. Hence, only small amounts were available. Iron's melting point is 1538°C; therefore, it did not melt in the furnaces being used prior to 1000 BCE.

Around 1000 BCE, metallurgists developed processes that enabled them to extract iron from minerals such as siderite (FeCO_3). A very hot charcoal fire is required for this chemical process to occur:



The charcoal burns with oxygen to form carbon monoxide (CO). The CO then reacts with FeO to form iron.



The iron is then hammered into shape while it is 'red hot' from the flame.

Once people could extract iron and shape it, they used it extensively for the production of weapons since it was harder than bronze.



Getty Images/De Agostini

Figure 2.5 ▲

Celtic iron axes and spears, from excavations at Illerup, Denmark

Why materials are mostly not found as elements

There are now known to be more than 114 elements, with 92 of these elements found in natural materials whereas more than 20 have now been produced using nuclear reactors and particle accelerators.

Elements are pure substances composed of one type of atom. The most stable elements in the periodic table are the noble gases. Except for helium, the atoms of these elements have an electron configuration with eight electrons in their valence shell, the most stable electron configuration. Atoms of other elements react to achieve this stable arrangement of electrons.

Ionic compounds

The simplest ions are formed when elements gain or lose electrons to achieve the same electron configuration as its nearest noble gas. Electrostatic forces of attraction occur between cations (positive ions) and anions (negative ions). The minerals used in the Bronze Age and Iron Age were examples of ionic compounds (Table 2.1). Chemists use chemical formulas to represent the ratio of atoms of each element present in a compound.

Table 2.1 Ionic compounds used in the Bronze Age and Iron Age

Mineral	Formula	Ions present
Malachite	$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$	Cu^{2+} , CO_3^{2-} , OH^-
Chalcopyrite	CuFeS_2	Cu^{2+} , Fe^{2+} , S^{2-}
Haematite	Fe_2O_3	Fe^{3+} , O^{2-}
Siderite	FeCO_3	Fe^{2+} , CO_3^{2-}

WOW

Legend of iron smiths

Legend has it that Wayland the Smith from Berkshire, UK, was commissioned by Merlin to make the sword Excalibur for King Arthur.



To learn about ionic bonding and compounds, refer to Chemistry sections 3.3 and 3.4 on pages 168 and 175.

Reactivity

Elements have different physical properties, such as melting point, density and hardness. They also have different chemical properties; that is, they react in different ways. Many elements react with water and oxygen in the atmosphere to form metallic oxides. This process is called **corrosion**.

Not all metals corrode at the same rate. These metals have different reactivities with oxygen. Iron corrodes quite quickly. Objects made from iron or alloys of iron such as steel corrode quite quickly and need to have a form of protection to minimise the corrosion. Aluminium does not corrode as quickly so is useful in window frames.



To learn about reactivity, refer to Chemistry section 4.2 on page 209.



Corbis/Gordon Wiltse/National Geographic Society

▲ Figure 2.6

Ships are made from alloys of iron, which corrode easily.



Alamy/Ernst Wlba

▲ Figure 2.7

Aluminium window frames do not corrode easily.

However, because most elements are reactive, there are many more compounds naturally present than elements. Minerals are being continuously formed, although it is a very slow process. Sometimes, these processes occur close to Earth's surface where water dissolves rocks, and crystals form over time. Limestone formations in caves are an example of this process. Limestone is mostly calcium carbonate (CaCO_3).



iStockphoto/markchenix

▲ Figure 2.8

Limestone formations in a cave are produced when water dissolves rocks and crystals of CaCO_3 form.



iStockphoto/plasticsteak1

▲ Figure 2.9

The granite in these kitchen benchtops formed deep under ground.

Sometimes these processes occur deep under ground where magma comes in contact with rock and melts it. Hence, as these materials react and cool over time, new minerals are formed. Therefore, most natural materials are compounds or mixtures of compounds. Granite has been formed in this way. It is a mixture of quartz, feldspar and other minerals. Granite is polished and used as benchtops in kitchens.

Natural materials

Many materials occur in nature. These are called natural materials. Materials such as wood and stone have been used to make houses. Originally, very little processing was done to materials such as wood. However, over time, more sophisticated structures have been built and the materials used have been protected by stains and varnishes.



Figure 2.10 ▲

The wood used to build both this log cabin and the more sophisticated wooden house is a natural material.

Properties

Each material has specific properties, which determine its use. Stone has often been readily available, but it is also generally very dense, which makes it heavy and difficult to move from place to place. Wood, however, is generally much less dense than stone and easier to move. Wood was a common material for building houses – it is strong, withstands rain and wind, and is easy to cut to required lengths and shapes. However, it could not be used for fireplaces because it is flammable. Stone was used for fireplaces and chimneys as it could withstand reasonably high temperatures.

Natural fibres such as cotton and wool were used for clothes and bed coverings.



Figure 2.11 ►

Stone is used for fireplaces because it can withstand high temperatures and is not flammable.



istockphoto/lenta



istockphoto/Byrdyak

▲ **Figure 2.12**

Cotton and wool are natural materials.

ACTIVITY 2.4

PROPERTIES OF NATURAL MATERIALS

Aim

To identify the properties of natural materials and relate them to their uses

What to do

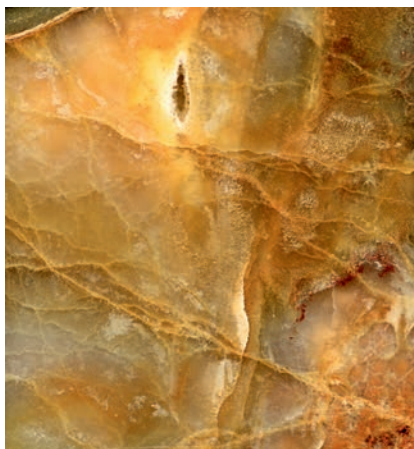
- 1 Research the properties of wool and cotton.
- 2 Discuss why wool and cotton are used for clothes and bed coverings.
- 3 Identify any limitations of wool and cotton and how this may affect their use for clothes and bed coverings.
- 4 Choose one other natural material, research its properties and relate these to its uses.

Uses

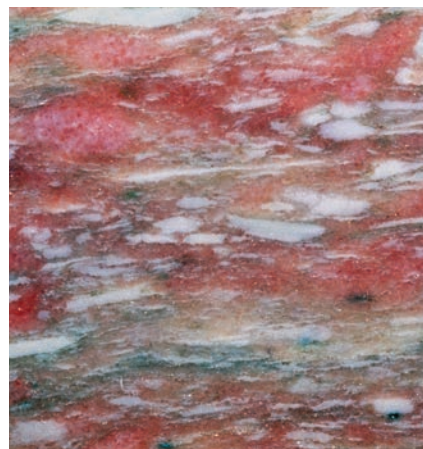
Some materials that occur in nature can be used with minimal processing. One example is marble, which is formed by the metamorphosis of sedimentary carbonate rocks such as limestone. The many different colours of marble are produced by impurities in the original limestone, as seen in Figure 2.13 on page 40. The stone is cut and polished before it is used for columns and benchtops.



Alamy/imagebroker



Getty Images/rusm



Science Photo Library/George Bernard

Figure 2.13 ▲

Marble contains impurities, which can make it white-grey, yellow or pink in colour.



To learn about covalent bonding and compounds, refer to Chemistry sections 3.5 and 3.6 on pages 178 and 185.

Covalent compounds

Covalent compounds are pure substances that are formed when atoms of two or more non-metals gain electrons to achieve the stable electron configuration of their nearest noble gas. **Hydrocarbons** (compounds that only contain carbon and hydrogen), such as alkanes and alkenes, are covalent compounds. These compounds are important for fuels.

Natural materials that need processing

Other natural materials, such as crude oil, need more processing before they can be used. Crude oil is a mixture of many different hydrocarbons. Crude oil is classed as a fossil fuel because it is formed by the decomposition of prehistoric living organisms. It takes millions of years to form.

Since crude oil is a mixture of hydrocarbons, it is processed using fractional distillation. Fractional distillation separates substances that have different boiling points. The substance that is most likely to form a vapour is said to be the most **volatile** substance. The substance with the lowest boiling point will evaporate first, then rise up into the fractionating column where it is cooled and collected.

INVESTIGATION 2.2

FRACTIONAL DISTILLATION

Fractional distillation uses the physical property of boiling point to separate mixtures of liquids. In this investigation, you will separate a mixture of ethanol and water because they have different boiling points.

What is your aim?

Write an aim that outlines the purpose of your investigation.

What will you need?

Figure 2.14 indicates some specialist equipment used to separate two liquids.

Research how you will need to modify the equipment to ensure that the mixture is heated gently and safely.

What equipment will you need so that you can identify the liquid that is collected first.

Research how you can test the first liquid to determine its identity.

Research how you can test the second liquid to determine its identity.

What are the risks?

What are the risks in doing this investigation?	How can you manage these risks to stay safe?
Chemicals could splash.	Wear safety glasses and wash your hands at the end of the investigation.
Glassware, especially the flask, will become hot.	Wait for equipment to cool before handling it.
Volatile mixtures could ignite or explode.	Move the Bunsen burner away from the boiling flask periodically to moderate the heat on the flask.
Ethanol is a potentially hazardous organic liquid.	Dispose of ethanol in an organic waste bottle.

In your write-up, add any more risks you can think of, as well as ways to manage them.

How will you carry out your investigation?

- 1 Figure 2.14 shows a diagram of the equipment required for this investigation.
- 2 Copy Figure 2.14 and annotate it to indicate the safety precautions that you've taken to ensure that the investigation will be performed safely.
- 3 Show your annotated diagram to your teacher and get it signed off before commencing your investigation.
- 4 Clearly describe the test you will perform to identify the first liquid that is collected.
- 5 Show this description to your teacher and get it signed off before performing the test.

What results will you collect?

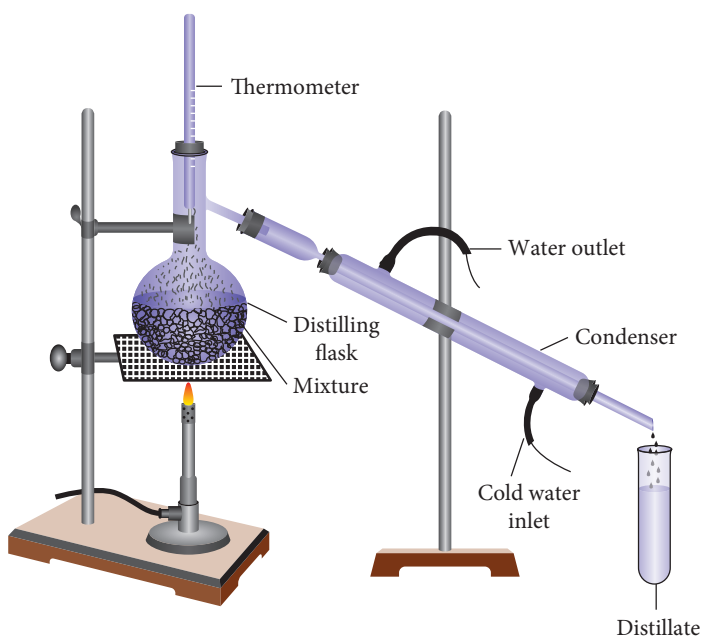
- 1 What information did you record when the first liquid was collected that may assist with identifying this liquid?
- 2 What information did you record when the second liquid was collected that may assist with identifying this liquid?
- 3 Describe and record your observations when testing each of the liquids.

What have you found?

- 1 At what temperature did the first liquid come through the water condenser?
- 2 What did you notice about the temperature recorded by the thermometer while the first liquid was going through the water condenser?
- 3 At what temperature did the second liquid come through the water condenser?

What can you conclude?

Justify your identification of each of the liquids with reference to its properties and the tests you performed.

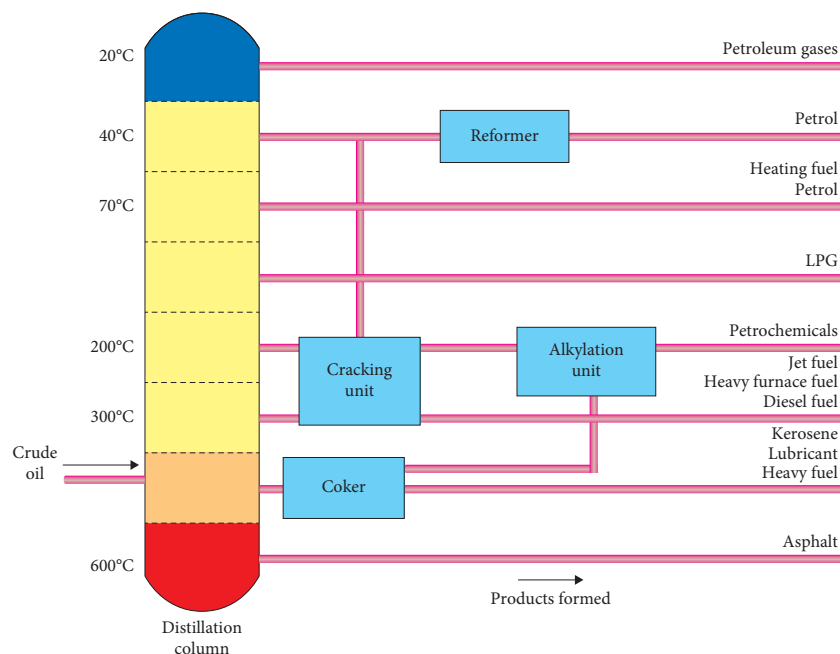


▲ Figure 2.14
Distillation apparatus

Oil refineries make the processing of crude oil economically viable and efficient. The crude oil is heated at normal atmospheric pressure to about 400°C in the distillation column, which has multiple exit points for volatile substances. The most volatile substances rise the highest in the column while the less volatile substances condense at lower levels in the column. This enables multiple fractions to be collected at the same time. Crude oil is continuously fed into the distillation column, which allows the process to continue.

Figure 2.15 ►

Distillation column in an oil refinery



To revise mixtures and methods of separation, refer to Chemistry section 2.3 on page 150.

ACTIVITY 2.5

CRUDE OIL FRACTIONS

Aim

To determine the hydrocarbons present in different crude oil fractions

What to do

Copy and complete the following table by referring to Figure 2.15 and other sources of information.

Fraction	Boiling point (°C) (may be a range)	Number of carbons in chain (may be a range)	Structural formula for one hydrocarbon in this fraction
Petroleum gases			
Jet fuel			
Petrol			
Kerosene			
Heating oil			
Diesel fuel			
Lubricants/waxes			
Asphalt			

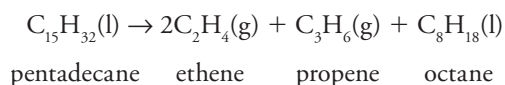
◀ **Figure 2.16**
An oil refinery



123RF/Dirk Ecken

Most fuels are not refined any further. However, some of the heavier fractions are processed further to increase the yields of the lighter fractions, which are the fractions in highest demand.

In Australia, some of the heavier fractions that have carbon chains with 15–25 carbon atoms, undergo **catalytic cracking**. Catalytic cracking is an example of a decomposition reaction where larger molecules are broken down to smaller molecules. This process uses a catalyst, a temperature of about 500°C and a moderately low pressure. The catalysts are **zeolites** (aluminosilicates), which are compounds that contain aluminium, silicon and oxygen. An example of a catalytic cracking reaction is:



The octane is used for petrol while the ethene and propene are very important starting materials for making plastics.

Fuels

Society uses a number of different fuels for different uses. Common fuels undergo **combustion** reactions to release the energy stored in the chemical bonds within these compounds. The energy is converted into useable forms.

In Australia, coal is the main source of energy for our power stations. Petrol and diesel are commonly used in our cars and propane is the main gas used for our barbecues. Petrol, diesel and propane are all hydrocarbons. These fuels are all classified as fossil fuels.

Hydrocarbons react with oxygen to produce carbon dioxide and water in a process called combustion. These exothermic reactions release large amounts of energy. Below is the combustion reaction for octane:



Petrol is a mixture of hydrocarbon chains of 5–12 carbons in length. A common component of petrol is octane (C_8H_{18}). Diesel is a mixture of much longer hydrocarbon chains, 10–24 carbons in length. A common component of diesel is hexadecane, which is commonly called cetane ($\text{C}_{16}\text{H}_{34}$).



To learn about hydrocarbons, refer to Chemistry section 3.8 on page 189.

The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)



To learn about energy and chemical reactions, refer to Chemistry section 4.3 on page 213.



To learn about moles, refer to Chemistry section 5.3 on page 235.



Properties and uses

Although petrol and diesel are both mixtures of hydrocarbons, they have slightly different properties since they are composed of different hydrocarbons. The density of diesel is 0.832 kg L^{-1} and the density of petrol is 0.745 kg L^{-1} . Diesel has a higher density than petrol because it consists of longer chain hydrocarbons that have a greater interaction between molecules than in petrol.

Both fuels release similar quantities of heat per kilogram of fuel that is combusted. One kilogram of diesel releases 43.1 MJ while the same mass of petrol releases 43.2 MJ. Carbon dioxide emissions are also important considerations when comparing fuels. Diesel releases $73.25 \text{ g CO}_2 \text{ MJ}^{-1}$ of energy produced, while petrol releases $73.38 \text{ g CO}_2 \text{ MJ}^{-1}$. Petrol engines require a spark plug to ignite the fuel, whereas diesel engines do not. Diesel engines compress the oxygen in the cylinder to high pressures and temperatures. The diesel is injected into the cylinder and ignites at these temperatures and pressures. Petrol is more flammable, likely to ignite, than diesel. A disadvantage of diesel is that at low temperatures, it becomes quite viscous, which changes its flow rate into the cylinder. This affects the fuel's performance. Additives have been developed that can be added to diesel fuel to overcome this problem. However, diesel engines are generally more fuel-efficient. A Mazda 6 with a diesel engine uses about 6.3 L per 100 km, whereas the same car with a petrol engine uses about 9.7 L per 100 km.

Figure 2.17 ▶

Diesel and petrol release similar amounts of energy and emit similar amounts of CO_2 .

AAP Image/Dave Hunt



To learn about combustion reactions, refer to Chemistry section 4.4 on page 221.

ACTIVITY 2.6

COMPARING PETROL AND DIESEL

Aim

To compare the properties of petrol and diesel

What to do

A person is trying to decide whether to buy a petrol car or a diesel car. Both cars have a 60 L fuel tank. The cost of diesel is currently 158.9 cents per litre while petrol is 131.9 cents per litre. The motorist travels about 15 000 km per year.

- 1 Determine which car has the more economical fuel by using the information about distance travelled, fuel cost, density of fuel, fuel economy and energy released per kilogram.
- 2 Research to determine which car is more environmentally friendly by using the information on the properties of the fuels, such as how much CO_2 they emit per year.
- 3 Which car would you recommend the person buy, and why?

Other useful natural materials

Australia has large deposits of minerals, and mining is a major industry and a major contributor to Australia's economy. Australia has significant reserves of ores for the metals aluminium, copper, gold, iron, lead, silver, zinc and uranium. These ores must undergo processing to extract these metals.

Uranium mining

Uranium is a naturally occurring actinide. Its three main isotopes are uranium-237, uranium-235 and uranium-234, which are radioactive and have long half-lives. Australia has 30–40% of the world's recoverable uranium. There are currently mining operations in South Australia and the Northern Territory and mining started in Western Australia in 2014. Mining of uranium is banned in Queensland. Both exploration and mining of uranium is banned in New South Wales and Victoria, so the extent of deposits in these states is unknown. Currently, Australia exports 10 000–14 000 tonnes each year. Australia only exports to countries that have signed agreements that specify that the uranium will only be used for peaceful purposes in civil nuclear power stations.

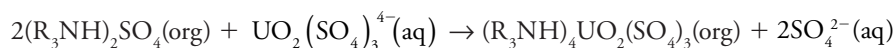
Extraction and processing

Extracting uranium from its ore and processing it so that it is suitable for use in a nuclear reactor is a complex process. Some of this process occurs near the mining operation whereas the final processing occurs at a refinery.

The two most common mineral sources of uranium are uraninite and pitchblende. Zircon, thorite and fluorite are generally also present so the extraction process must both isolate and purify the uranium compounds from the raw ore. Uraninite and pitchblende both have the chemical formula UO_2 ; they also contain the greatest percentage composition of uranium, about 88%. Other minerals, such as carnotite ($\text{K}_2(\text{UO}_2)_2(\text{VO}_4)_2 \cdot 3\text{H}_2\text{O}$) have a lower percentage composition of uranium. Hence, a much greater mass of the ore, carnotite, must be processed to produce the same mass of uranium as that produced by pitchblende or uraninite.

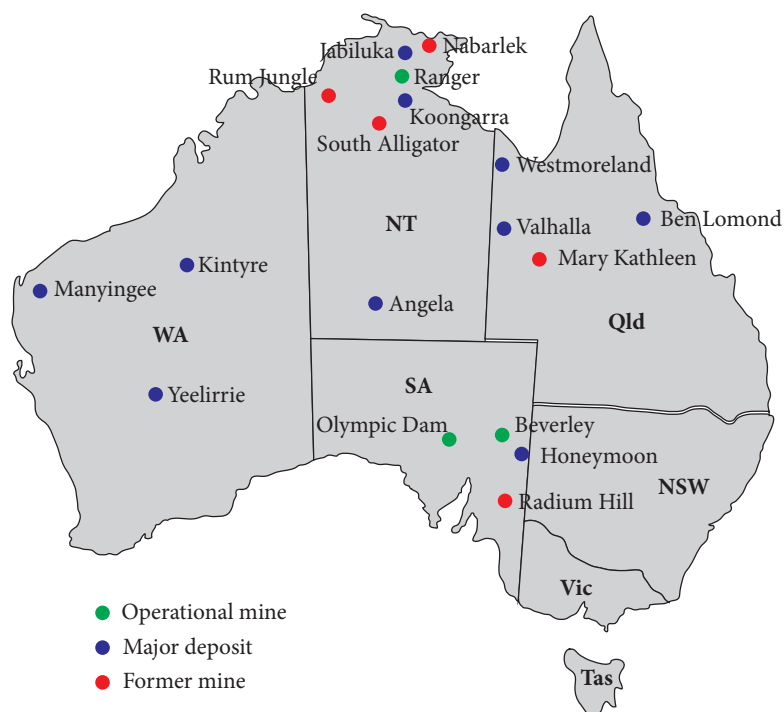
There are a couple of ways of extracting uranium. The Ranger and Olympic Dam mines in the Northern Territory and South Australia use the method in the following paragraphs. The mine at Beverley in South Australia uses a slightly different method.

The ore is crushed and ground to reduce the particle size. Then it is put through a leaching process to make the uranium salt soluble. Depending on the nature of the ore, this process could use an alkaline leaching agent or an acidic leaching agent. At the Ranger and Olympic Dam mines, an acidic leaching agent, H_2SO_4 , will be used to form $\text{UO}_2(\text{SO}_4)_3^{4-}$ (uranyl sulfate ion). Solvent extraction then both concentrates and purifies the uranium. An organic kerosene solvent containing tertiary amines is added to the uranyl sulfate ion according to the following equation:



AUSTRALIAN ATLAS OF MINERALS RESOURCES, MINES AND PROCESSING CENTRES

Create a movie or vodcast about an Australian metal, including location of deposits, processing and uses.



▲ Figure 2.18
Australian uranium deposits and mines

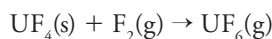
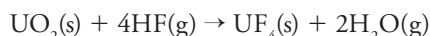
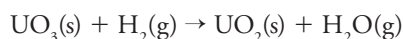


iStockphoto/John Camemolla

Figure 2.19 ▲

This uranium mine is at Ranger Mine, Northern Territory.

heating leads to the production of pure UO_3 , which undergoes the following reactions to form uranium hexafluoride (UF_6).

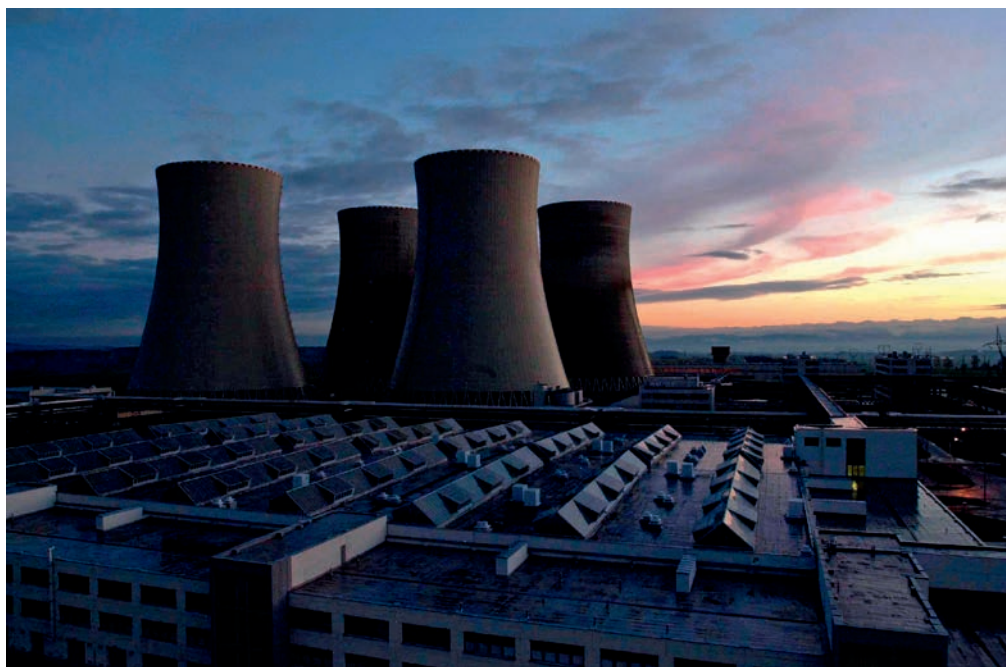


Mining companies need to understand the chemistry of the processes so they can maximise the financial return on their investment. They can maximise the yield of the required product, and minimise the quantities of reagents used and waste products produced. According to the equations above, 1 mole of $\text{UO}_3(\text{s})$ produces 1 mole of $\text{UF}_6(\text{g})$. During this process, 1 mole of $\text{H}_2(\text{g})$, 4 moles of $\text{HF}(\text{g})$ and 1 mole of $\text{F}_2(\text{g})$ are used and 3 moles of $\text{H}_2\text{O}(\text{g})$ are produced. Mining companies convert these quantities from moles to masses because they generally quote figures per tonne of $\text{UO}_3(\text{s})$ used or per tonne of $\text{UF}_6(\text{g})$ produced.

The uranium hexafluoride, UF_6 , then undergoes gaseous diffusion, to separate uranium isotopes to produce enriched uranium that can be used in nuclear reactors. This process involves increasing the percentage of uranium-235 from its natural levels of about 0.7% to about 4%. The enriched uranium is then converted to uranium dioxide pellets, UO_2 , for use in the nuclear reactors, such as that shown in Figure 2.20.



To learn about mass and moles, stoichiometry and the law of conservation of mass, refer to Chemistry sections 5.4 and 5.6 on pages 239 and 248.



Corbis/David Veis/epa

Figure 2.20 ►

The Temelin nuclear power plant in the Czech Republic

ACTIVITY 2.7

PROCESSING URANIUM

Aim

To present information about uranium mining in a flowchart

What to do

- 1 Present the information about the extraction and refining of uranium in a flowchart. This can be done manually or by using a program such as Gliffy.
- 2 Identify the physical processes in the extraction and refining of uranium.
- 3 Identify the chemical processes in the extraction and refining of uranium.

Social, economic and cultural considerations

Uranium mining is a sensitive issue. Australia's uranium mines are located in remote areas near Aboriginal communities. The Ranger mine, near Jabiru (about 260 km east of Darwin) in the Northern Territory, is on land owned by the Mirarr Aboriginal people. It is surrounded by the Kakadu National Park, which is a World Heritage listed area due to its extensive Indigenous art sites and internationally significant wetlands. The Mirarr people have lived on this land for more than 50 000 years. Dreaming tracks cross the land and there are many sacred sites, including the Ranger mineral lease.

The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)



◀ **Figure 2.21**
Wetlands, Yellow Water Lagoon, Kakadu National Park

The Ranger Project Area was exempt from the Aboriginal Land Rights Act so the Aboriginal people did not have a say on whether mining occurred. The government considered the benefits to the Australian economy in terms of jobs generated and export dollars received to outweigh the concerns of the traditional landowners. However, the Mirarr people did have a say on the conditions under which the mining occurred. There was considerable dispute between the Northern Land Council, with whom the Australian Government and the mining company, Energy Resources of Australia (a subsidiary of Rio Tinto Pty Ltd), were negotiating, and the Mirarr people, the traditional owners of the land. The Ranger mine commenced operation in 1980. Over the last 30 years, over 100 000 tonnes of uranium oxide have been produced during open-cut mining operations at Ranger – one of only three uranium mining operations in the world to reach this figure. These operations ceased at the end of 2012. In January 2013, a 14-year battle to renegotiate the original agreement between Energy Resources of Australia (ERA) and the Gundjeihmi Aboriginal Corporation, who acted on behalf of the Mirarr people, was settled. This agreement will lead to ERA paying increased royalties to the Mirarr people. They will also set up a sustainability fund for the Mirarr people. ERA is planning to continue mining at Ranger; however, they are moving to underground mining operations.

Evaluating media and scientific claims

Much has been and continues to be written in the media about the Ranger uranium mine. Most of the articles are about the agreement between the Mirarr people and ERA or environmental issues related to the mining operation, specifically water management.

ACTIVITY 2.8

EVALUATING CLAIMS ABOUT THE RANGER URANIUM MINE

Aim

To evaluate claims made about issues related to the Ranger uranium mine

What to do

- 1 Find three articles about the Ranger uranium mine from different organisations; for example, the Australian Broadcasting Corporation (ABC), Environment Centre NT, Energy Resources of Australia (ERA) and CSIRO.
- 2 For each article:
 - a identify the claim(s) made.
 - b outline the implication(s) for each claim.
 - c research and identify any scientific evidence available that may support or refute the claim.
- 3 For one of the claims that you have researched, present your findings in a format of your choosing; for example, iMovie, Blabberize, podcast, cartoon, PowerPoint.

Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)



Finding new materials

Many materials are abundantly available and easy to extract or process. These natural materials have been used for thousands of years. Some materials have only been discovered or synthesised more recently. Other materials are only present in very small quantities so specialised equipment is needed to find and extract them. Other substances are only produced when scientists understand the chemical and physical properties of the material and therefore can see the possibilities in terms of manipulating the material or reacting it with another material to form a new material with different properties.

Rare earth elements

The rare earth elements, more commonly known as the lanthanoids, are elements 58–71 in the periodic table; that is, from cerium to lutetium. The International Union of Pure and Applied Chemistry (IUPAC) classifies 17 elements as rare earth elements – the lanthanoids and lanthanum, yttrium and scandium. Although they are called rare, they are actually reasonably abundant. Thulium, which is the least abundant, is more abundant than gold. However, their deposits are fairly localised; hence, the reference to ‘rare’. Originally, they were discovered in Sweden, but there are now known deposits on every continent except Antarctica.

About 2000 tonnes of mine tailings are produced for each tonne of rare earth element mined. The mine tailings contain radioactive thorium. China is currently the world supplier of rare earth elements; it has significantly decreased its production over the last couple of years, which has led to an increase in the price of these elements. Other countries, such as the USA and Australia, are currently determining whether it is economically viable to develop their rare earth deposits and the infrastructure necessary to process these elements.

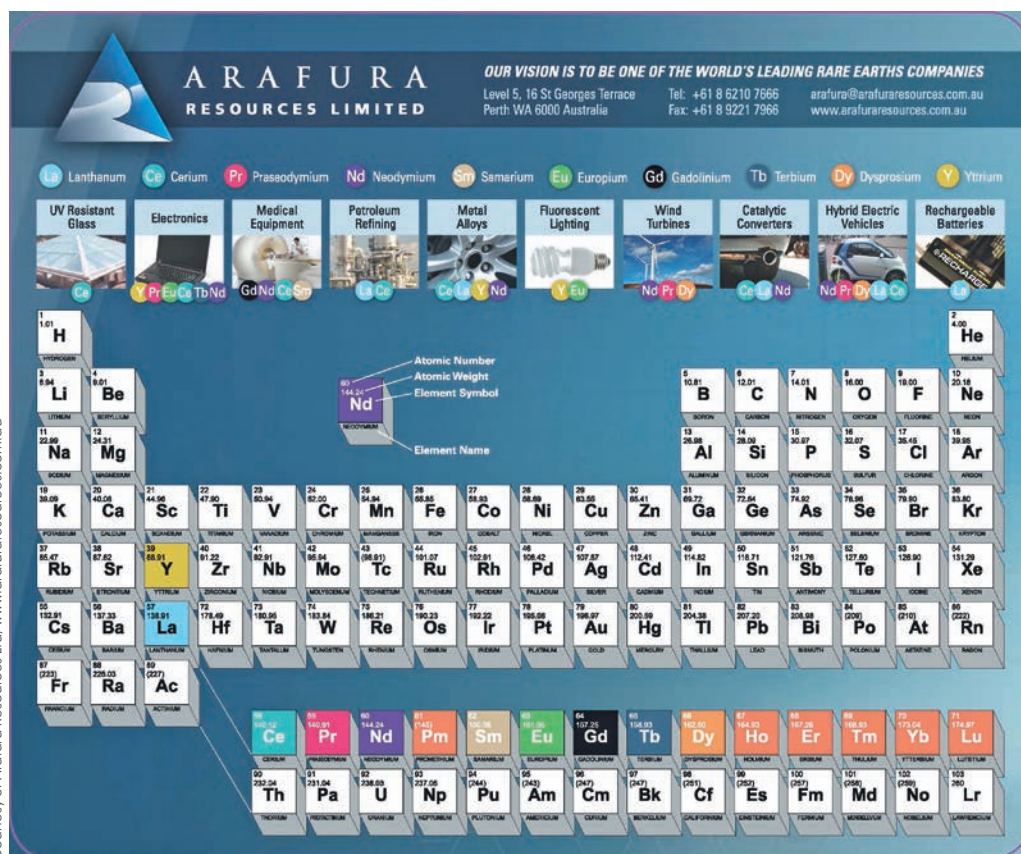
WOW

Magnets

The magnetic fields of neodymium magnets are 10–50 times stronger than those of iron-based magnets. Therefore, if we used iron-based magnets in our mobile phones, then they would need to be 10–50 times larger to exhibit the same size magnetic field. Imagine how ‘chunky’ our mobile phones would be!

Extraction

Deposits of rare earth elements are fairly localised and most deposits contain mixtures of rare earth elements, which contain relatively small amounts of individual rare earth elements. This makes their extraction an expensive process.



◀ Figure 2.22
Rare earth elements –
properties and uses



AN INTRODUCTION TO THE RARE EARTH ELEMENTS

Watch this video from Molycorp Minerals that shows uses for some of the rare earth elements.



THE GLOBAL RACE FOR RARE EARTHS

Watch this video from the Bureau for International Reporting about some of the issues related to mining, supply and uses of rare earth elements.

The oxides of the rare earth elements all have very similar high melting points. Not only do the oxides have similar properties, but the rare earth elements themselves have similar melting points and they also have the same stable valency, 3+. All of this means that it is extremely difficult to extract rare earth elements.

Although the ions of the rare earth elements have the same charge, they have different ionic radii since the ratio of protons to electrons is slightly different. This, in turn, leads to different-sized forces of attraction between the nucleus and the valence shell electrons. Generally, as the number of protons increases, the radius of the ion decreases.

The breakthrough technology for extracting rare earth elements was the invention of ion-exchange chromatography. This process allows for the collection of substantial quantities of pure rare earth elements.

During ion-exchange chromatography, the cations of the various rare earth elements bind to the **chelating ligand** (a covalent substance whose molecules can form several bonds with each single metal ion) in the **eluting** solution to different degrees. Hence, they take different amounts of time to be washed from the chromatographic column. The smaller ions bind more tightly to the chelating ligands and, therefore, they move through the chromatographic column in the shorter time.

Uses

Rare earth elements have many uses. New hybrid vehicles such as the Toyota Prius use nickel–lanthanum hydride batteries. These batteries are more powerful than the lead–acid batteries used in ordinary vehicles and yet they are also smaller in size. Each Prius contains about 4.5 kg of lanthanum.

Europium was used in cathode ray tube televisions for the red phosphors that produced the vibrant red colour in pictures. This led to the introduction of colour television in the late 1960s and early 1970s. Although these televisions are being phased out and replaced by LED/LCD (light-emitting diode/liquid-crystal display) and plasma televisions, europium is being used in one of the colours that are mixed to produce the white LED-based lights.

However, generally only small quantities of rare earths are added to other materials. Erbium is added in small amounts to optical fibres to amplify the light pulses; this means that the signal maintains its strength and hence can travel greater distances. Rare earth elements are at the forefront of many environmentally friendly technologies.

ACTIVITY 2.9

RARE EARTH ELEMENTS

Aim

To research the properties and uses of a rare earth element

What to do

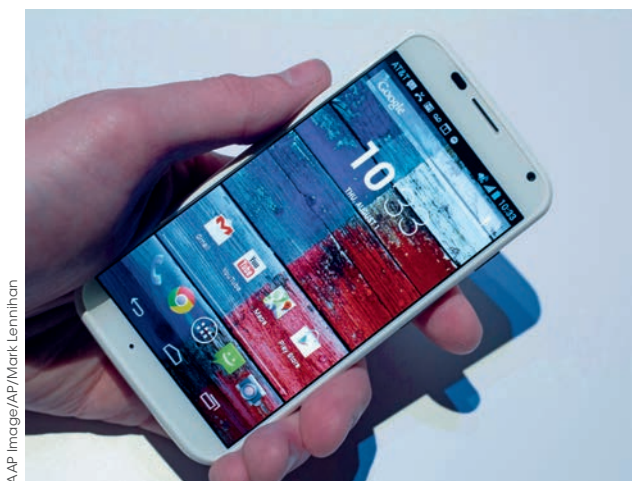
Choose a rare earth element.

- Identify its properties.
- Describe at least one use and relate it to its properties.
- Outline its abundance.
- Discuss any implications for society if its supply were to be interrupted or terminated.

Looking after our resources

We live in a consumer world. We have more goods and appliances than ever before. All of these goods are packaged. Hence, there is high demand for our resources. Scientists are continually trying to develop new materials or new processes for production of existing materials that have less impact on the environment. Some examples are recycling existing materials and the development of biofuels.

Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability (ACSCH015 and ACSCH054)



AAP Image/AP/Mark Lennihan



Corbis

▲ Figure 2.23

Consumer goods such as smart phones and home cinemas require a lot of resources and energy to produce.

Social, economic and ethical considerations

Society has started to become more aware of the impact of consumerism, not just in terms of environmental effects but also the social and ethical considerations. Society's demand for affordable and luxury goods means that companies are trying to produce them for the cheapest possible prices. This means that many of the manufacturing plants are in countries such as Thailand, China, India and Bangladesh, where wages are low.

Our consumer world has had many positive impacts on our lives. However, many people are concerned about the changing nature of society – for example, people sitting inside watching television and playing computer and video games rather than participating in more active and communal pastimes such as sports.



ABC NEWS BREAKFAST – GET IT RIGHT ON BIN NIGHT

Watch this interview with a spokesperson from Sustainability Victoria on disposal of household waste. It indicates that people 15–30 years of age are least likely to recycle.

ACTIVITY 2.10

ETHICAL CONSIDERATIONS OF CONSUMERISM

Aim

To compare the working and living conditions of workers in Australia and in a country from which we get consumer goods

What to do

- Choose a type of manufacturing industry, such as the electronics, car or clothing industry.
 - For workers in that industry in Australia, outline their:
 - working conditions; for example, number of hours per day, any loadings such as sick leave, pay scales, holidays.
 - living conditions; for example, housing, appliance ownership.
- Choose a country from which Australia imports goods in that manufacturing industry.
 - For workers in that industry in that country, outline their:
 - working conditions; for example, number of hours per day, any loadings such as sick leave, pay scales, holidays.
 - living conditions; for example, housing, appliance ownership.
- Discuss how our standards of living are affecting the standards of living in other countries.
- Justify whether people living in Australia should expect a higher standard of living than people in other countries.
- Create an advertisement to make your peers more aware of the working conditions of workers in other parts of the world.

ACTIVITY 2.11

SOCIAL CONSIDERATIONS OF CONSUMERISM

Aim

To analyse some of the social impacts of living in a material world

What to do

- Research and justify the positive impacts on society of living in a material world.
- Research and justify the negative impacts on society of living in a material world.
- Conduct a class debate on the topic 'People who lived in the 1950s had a better quality of life than people living in the 2010s'.



Figure 2.24 ▲

Changing times: the shape, purpose, materials and technology of cars have changed over time.

Recycling

Materials from the earth are limited resources. As the resource becomes more scarce, the cost increases – it's all about supply and demand. Companies have to weigh up the options of whether it is more economically viable to mine for metals such as aluminium or to recycle existing stock.

Aluminium used to be a very expensive metal, so expensive that the Washington Monument in Washington, DC, USA, was given an aluminium tip to symbolise its value. However, in about 1886, a cheaper method was developed to extract aluminium from bauxite. The cost of aluminium dropped and the metal was used extensively for wrappings, window frames and soft-drink cans.

The extraction of aluminium from bauxite is still a very energy-intensive process. Recycling aluminium only uses 5% of the energy required to mine, extract and purify it from bauxite. To produce 1 kg of aluminium requires:

- 200 MJ of energy to mine, extract and purify it from natural ores, such as bauxite
- 7 MJ of energy from recycled material.



Alamy/National Geographic Image Collection



HOWSTUFFWORKS - RECYCLING ALUMINIUM

Watch this video about how aluminium is recycled.

◀ Figure 2.25

The Washington Monument was completed in 1885 at a time when aluminium was very expensive to produce.

ACTIVITY 2.12

MINING VS RECYCLING ALUMINIUM

Aim

To compare the chemical and physical processes in the production of aluminium from mining and recycling

What to do

- Research the physical and chemical processes involved in extracting aluminium from bauxite, including the reaction conditions.
 - Present the information as a flowchart, either hand drawn or created by using a program such as Gliffy.
- Research the physical and chemical processes involved in recycling aluminium.
 - Present the information as a flowchart, either hand drawn or created by using a program such as Gliffy.
- Refer to your flowcharts to explain why two-thirds of all aluminium mined is still in use today.

Biofuels

There are four main areas of focus in terms of biofuels for cars – ethanol, biodiesel, biogas and green gasoline. Biodiesel can be used with existing diesel engines, although currently in Australia only blends that have 5–20% biodiesel are available. They are labelled as B5 or B20. Green gasoline can be used with existing petrol engines, although this is not currently available in Australia. Compressed biogas can be used in motor vehicles. Although this is also not available in Australia, it is used in Sweden, Switzerland and Germany for cars, trucks and trains. In Australia, only ethanol is currently used as a fuel additive – E10 is a mixture of unleaded fuel and 10% ethanol.

Scientific knowledge can be used to develop and evaluate projected economic, social and environmental impacts and to design action for sustainability (ACSCH015 and ACSCH054)



WOW

First diesel engine

Rudolph Diesel, who invented the first diesel engine, designed it to run on peanut oil.

WOW

Power from the people

In 1895, the city of Exeter, in the UK, used gas produced from sewage to power its street lamps.



TURNING ALGAE INTO OIL THE NASA WAY

Watch this video on cutting-edge research being carried out on the production of biofuels.



To learn about biofuels, refer to Chemistry chapter 4 'Biofuels' on page 225.

Green gasoline is petrol produced from plants. Petrol is a hydrocarbon – it contains only carbon and hydrogen. Plants contain cellulose, a carbohydrate consisting of many glucose molecules, which contains carbon, hydrogen and oxygen. In petrol, the carbon–hydrogen bonds are broken and carbon–oxygen and hydrogen–oxygen bonds are formed. Glucose, $C_6H_{12}O_6$, already contains some oxygen, so some of the carbon–oxygen and hydrogen–oxygen bonds are already formed; hence, less energy is released when combustion occurs than with a similar-size hydrocarbon such as cyclohexane, C_6H_{12} . Approximately 3 kg of glucose would produce the same amount of energy as 1 kg of cyclohexane.

US scientists at the University of Massachusetts Amherst, Massachusetts, have successfully developed a process for stripping oxygen from glucose to form hydrocarbons similar to cyclohexane. However, the challenge is to make it economically viable on a larger scale. Virent Energy Systems in Madison, Wisconsin, are working on developing this process. The other issue is the source of cellulose, which is currently the waste material from agriculture and forestry, such as sawdust and the stalks left after the cobs of corn have been picked.

The largest supplier of ethanol as a fuel additive in Australia is the Manildra Group. Their source of ethanol is wheat. They process over 1 million tonnes of wheat each year for industrial uses, including the production of ethanol. Their flour mill is located between Orange and Parkes in western New South Wales, about 300 km west of Sydney. Their ethanol plant is located in Nowra, on the south coast of New South Wales, about 160 km south of Sydney.

Biogas is commonly used to refer to the gas that is produced from the decomposition of organic matter in the absence of oxygen; that is, under anaerobic conditions. The gases that are released are a mixture of methane (60–70%) and carbon dioxide (30–40%) with very small amounts of other gases such as hydrogen, nitrogen and hydrogen sulfide. Since biogas can be combusted to release energy, it can be used as a fuel. The composition of biogas varies with the biodegradable material used for its production. Biodegradable materials such as sewage, waste in landfill sites, waste in septic tanks and manure can all be used for the production of biogas. Many of Australia's waste recovery companies have developed pilot plants for biogas production at landfill sites or sewage treatment plants. The biogas at these sites is released into the atmosphere if the site has not been engineered to collect the gas.

Small-scale digestion plants have been set up in over 2 million households in India and several hundreds of thousands of households in Pakistan to produce 'gobar gas', which is produced from cow manure. It provides households with gas for cooking.

In Australia, biodiesel is produced from vegetable oils such as canola oil, animal fats and even recycled vegetable oil. There is also research into using algae as the source of vegetable oils for biodiesel. The advantage of using algae is the potential for high yields per hectare. Algae could produce 100 000 L of oil per hectare, while the next best yield is 5000 L per hectare from palm oil. The other advantage of algae is that prime agricultural land would not be required for its production.

Properties, uses and comparison to fossil fuels

Diesel is a mixture of different hydrocarbons in different proportions; therefore, its properties differ slightly depending on its source, which could be from fossil fuels or the biodiesel sources of vegetable oils and animal fats.

Biodiesel is a safer fuel than diesel because it has a higher **flashpoint** than diesel. The flashpoint is the temperature at which there is enough vapour present to ignite with oxygen when an ignition source is present. Biodiesel is also better for the environment. It is almost carbon neutral since CO_2 is taken out of the atmosphere to produce the plant sources for the biodiesel and returned to the atmosphere when the biodiesel is used. Some extra CO_2 is released into the atmosphere during the production of biodiesel from the vegetable oils and animal fats. However, diesel from fossil fuels does not take CO_2 from the atmosphere since it has taken millions of years to be produced, but it does release CO_2 into the atmosphere during the refinery process and when the diesel is used. Biodiesel also has a lower energy content than diesel from fossil fuels: it releases about 10% less energy.

Ethanol releases more than 30% less CO_2 into the atmosphere than petrol. This means that there are fewer environmental concerns about the use of this fuel. However, ethanol only releases about 60% of the energy that is released by petrol during combustion. This means that considerably more ethanol would be needed to provide the same amount of energy for use in a car. In Australia, E10 is a mixture of unleaded fuel with 10% ethanol. Therefore, more E10 would be used to produce

the same energy as unleaded fuel without ethanol added. However, the difference is less than if the vehicle was using ethanol alone.

Biogas is a mixture of gases but the major component is methane. Methane is also available as the major component in natural gas from reserves off the coast of north-west Western Australia, the Otway and Gippsland regions of Victoria and smaller basins in the Northern Territory, Queensland and South Australia. Natural gas and biogas can both be compressed and hence have the potential to be used in vehicles.

The greatest issue for the use of all biofuels is the cost of production when compared to production from fossil fuels. To reduce these costs, at present, ethanol is blended with petrol and biodiesel is blended with diesel.

ACTIVITY 2.13

BIOFUEL PRODUCTION AND USE IN AUSTRALIA

Aim

To describe the production process for one type of biofuel in Australia and compare it to one of the fossil fuels

What to do

- 1 Choose one biofuel being produced in Australia – biogas, ethanol or biodiesel. Describe how it is produced, including relevant diagrams.
- 2 Describe some of the properties of your chosen biofuel, either in absolute terms or by comparison with an identified fossil fuel.
- 3 Develop an advertisement to promote awareness of the availability of your chosen biofuel.

Biomimicry – learning from nature

Scientists are always looking to develop new materials that perform better than existing materials or are used for new purposes. Geckskin is a new material developed by scientists at the University of Massachusetts Amherst. Researchers looked at how geckos were able to cling to very smooth surfaces instead of falling off. Geckskin is a material that adheres very strongly to smooth surfaces such as glass. A 10 cm by 10 cm piece of the material can support the weight of about four people hanging from it. Although it adheres very strongly, it can be removed from the glass very easily and without leaving any residue. Possible uses could be to replace ladders in rescue situations.

Another new material is Shrilk. This biodegradable material has been developed by scientists at the Wyss Institute for Biologically Inspired Engineering at Harvard University, Massachusetts, USA. Shrilk is made from chitosan, a polymer developed from chitin, which is the substance in the exoskeletons of crustaceans and insects, and fibroin, a protein derived from silk. The researchers looked at insects, not just for the materials but also for how to put the materials together. In insects, the chitin and fibroin are layered, so researchers layered the chitosan and fibroin to produce a material that is thin, clear, flexible and strong. A piece of Shrilk is lighter but stronger than a comparably sized piece of aluminium. Due to the availability of chitosan and fibroin, Shrilk is relatively cheap to manufacture and, importantly, it is biodegradable under conditions at landfill sites. The raw materials for making Shrilk have already been approved by the US Food and Drug Administration for use on humans. Potential uses of Shrilk include plastic bags, sutures that would dissolve over time and possibly as a scaffold for regenerating tissue.



▲ **Figure 2.26**
Gecko feet were the inspiration for a new material called Geckskin, which held up to four people while adhering to a smooth wall.

CHAPTER SUMMARY

- Humans have been looking for materials with specific properties to use for specific purposes for thousands of years. This began with very primitive materials in the Stone Age, and then became more advanced as processes were developed to extract metals from ores during the Bronze Age and Iron Age.
- Some natural materials are useful with minimal processing; for example, marble, wool and cotton. However, many materials need to be processed to be useful; for example, fuels. These materials can undergo physical and chemical processes to make them useful.
- Most elements are very reactive so are found naturally in compounds. The more reactive elements require more technologically advanced methods to extract them from their compounds.
- Mining is a major industry in Australia, which involves large-scale operations of physical and chemical processes to extract the metal from its ore. Uranium mining is one such example. Before a mining operation is established, many economic, social, cultural, environmental and ethical issues need to be considered.
- A deep understanding of the properties of different elements has enabled processes to be developed for isolating specific rare earth elements from ores. This has enabled the development of new technologies, such as stronger magnets and batteries for hybrid vehicles.
- Humans are becoming more conscious of their impact on the environment. Recycling is one way of minimising this impact. Chemists are also developing new materials from more renewable sources; for example, crops being used to produce green gasoline and biodiesel. Renewable fuels must have similar properties and uses as existing fuels from non-renewable sources.
- Chemists are continually developing new materials. They are analysing materials in nature to develop new materials, such as Geckskin and Shrilk.

CHAPTER GLOSSARY

alloy a mixture of two or more elements, one of which must be a metal

catalytic cracking the chemical process of breaking longer-chain hydrocarbons into shorter-chain hydrocarbons

chelating ligand a covalent substance whose molecules can form several bonds with each single metal ion

combustion a reaction with oxygen to form the oxides of each of the elements present; with adequate oxygen, combustion of a hydrocarbon will produce carbon dioxide and water

corrosion a chemical reaction in which a metal degrades in the presence of oxygen and water to form the oxide of the metal

covalent compound a compound composed of atoms of at least two different non-metals chemically combined in definite proportions

eluting separating the parts of a mixture by using the property that they travel through a solvent at different rates

environmental footprint a measure of the impact of humans on Earth's ecosystems and resources

flashpoint the lowest temperature at which a fuel has enough vapour present to ignite in air

hydrocarbon a chemical compound composed only of carbon and hydrogen

volatile can change from a liquid to a gas (vaporise) easily

zeolite an aluminosilicate mineral; i.e. contains Al, Si and O